

Categorical Data Descriptive Statistics

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Data

We will use the data which contains artificial supermarket transaction data. The data frame consists of 16 variables. We shall illustrate the ways to compute the summary statistics and to visualize the categorical data.

```
library(readxl)
supermarket <- read_excel("D:/BSDS/Statistics 3/SupermarketTransactions.xlsx",sheet=2)
head(supermarket[, c(3:5,8:9,14:16)])
```

```
## # A tibble: 6 x 8
##   `Customer ID` Gender `Marital Status` `Annual Income` City `Product Category`
##         <dbl> <chr>   <chr>           <chr>           <chr> <chr>
## 1         7223 F       M             $30K - $50K     Los ~ Snack Foods
## 2         7841 M       M             $70K - $90K     Los ~ Vegetables
## 3         8374 F       M             $50K - $70K     Brem~ Snack Foods
## 4         9619 M       M             $30K - $50K     Port~ Candy
## 5         1900 F       S             $130K - $150K   Beve~ Carbonated Bevera~
## 6         6696 F       M             $10K - $30K     Beve~ Side Dishes
## # i 2 more variables: `Units Sold` <dbl>, Revenue <dbl>
```

Frequencies

We need to produce the **contingency table** which calculates the counts for each combination of categorical variables.

For instance, we may want to get the total count of female and male customers:

```
table(supermarket$Gender)
```

```
##
##      F      M
## 7170 6889
```

If we want to understand the number of married and single females and male customers we can produce a cross classification table:

```
table(supermarket$`Marital Status`, supermarket$Gender)
```

```
##
##      F      M
## M 3603 3264
## S 3567 3625
```

We can also produce multidimensional tables based on three or more categorical variables. In this case we assess the count of customers by marital status, gender, and location:

```
table1 <- table(supermarket$`Marital Status`, supermarket$Gender,
                 supermarket$`State or Province`)
ftable(table1)
```

```
##          BC   CA   DF Guerrero Jalisco   OR Veracruz   WA Yucatan Zacatecas
##
## M F   190  639  188      77      15  510      142 1166      200      476
##  M   197  692  210      94       5  514      108 1160      129      155
## S F   183  685  175     107     30  607      125 1134      164      357
##  M   239  717  242     105     25  631       89 1107      161      309
```

Proportions

We can also produce contingency tables that present the proportions (percentages) of each category or combination of categories. The following reproduces the previous tables but calculates the proportions rather than counts:

```
table2 <- table(supermarket$Gender)
prop.table(table2)
```

```
##
##          F          M
## 0.5099936 0.4900064
```

```
table3 <- table(supermarket$`Marital Status`, supermarket$Gender)
prop.table(table3)
```

```
##
##          F          M
##  M 0.2562771 0.2321644
##  S 0.2537165 0.2578420
```

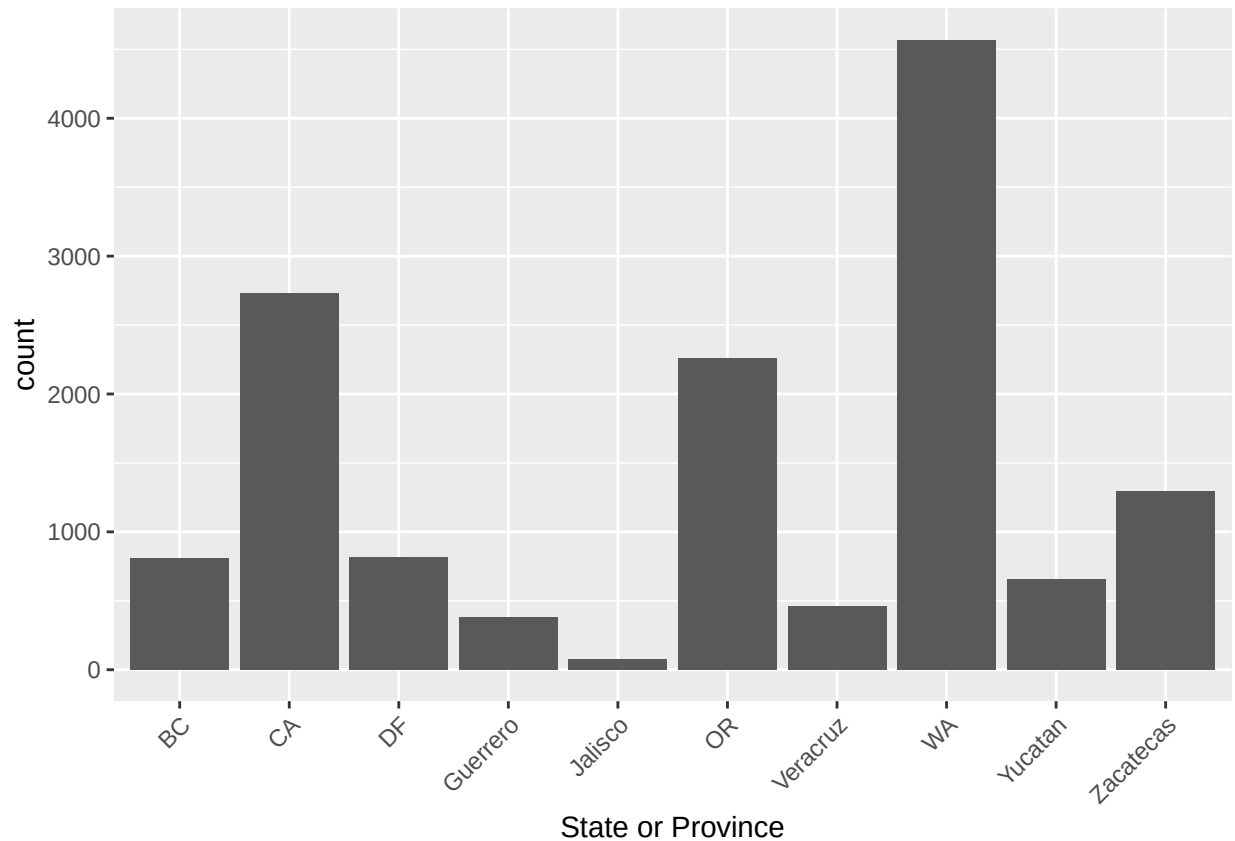
```
ftable(round(prop.table(table1), 3))
```

```
##          BC   CA   DF Guerrero Jalisco   OR Veracruz   WA Yucatan Zacatecas
##
## M F  0.014 0.045 0.013   0.005   0.001 0.036   0.010 0.083   0.014   0.034
##  M  0.014 0.049 0.015   0.007   0.000 0.037   0.008 0.083   0.009   0.011
## S F  0.013 0.049 0.012   0.008   0.002 0.043   0.009 0.081   0.012   0.025
##  M  0.017 0.051 0.017   0.007   0.002 0.045   0.006 0.079   0.011   0.022
```

Visualization

Bar charts are most often used to visualize categorical variables. Here we can assess the count of customers by location:

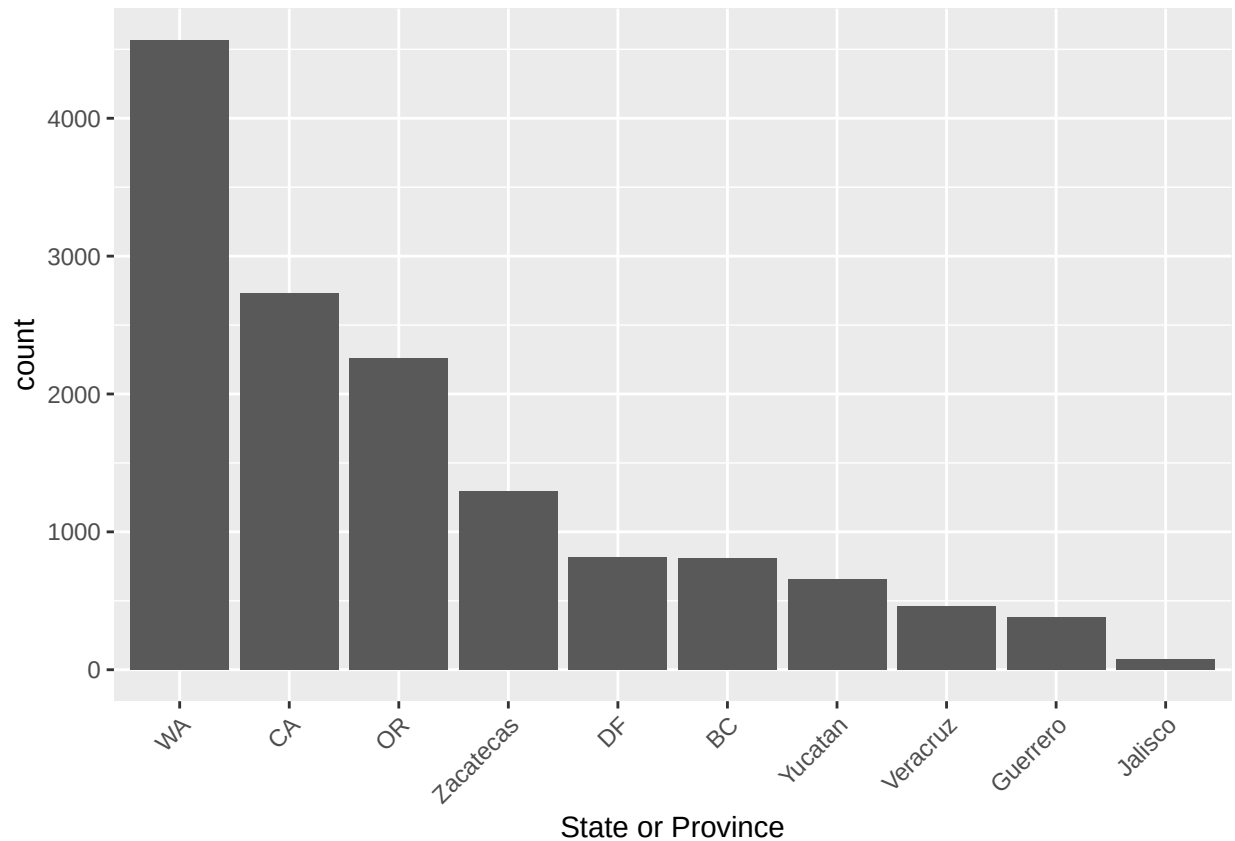
```
library(ggplot2)
ggplot(supermarket, aes(x = `State or Province`)) +
  geom_bar() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



To make the bar chart easier to digest we can reorder the bars in descending order.

```
# re-order levels
reorder_size <- function(x) {
  factor(x, levels = names(sort(table(x), decreasing = TRUE)))
}

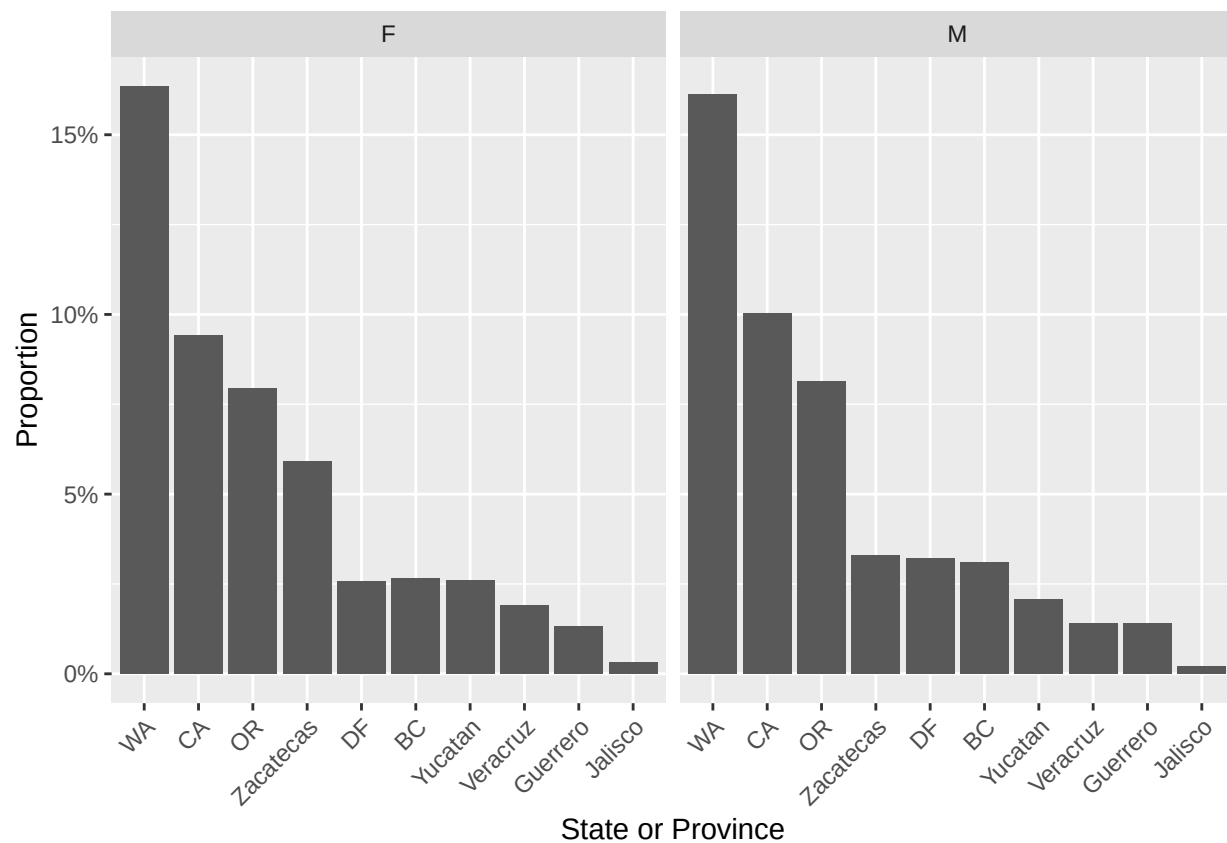
ggplot(supermarket, aes(x = reorder_size(`State or Province`))) +
  geom_bar() +
  xlab("State or Province") +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



We can also create contingency table-like bar charts by using the `facet_grid()` function to produce small multiples. Here I plot customer proportions across location and by Gender.

```
ggplot(supermarket, aes(x = reorder_size(`State or Province`))) +
  geom_bar(aes(y = (..count..)/sum(..count..))) +
  xlab("State or Province") +
  scale_y_continuous(labels = scales::percent, name = "Proportion") +
  facet_grid(~ Gender) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

```
## Warning: The dot-dot notation (``..count..``) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(count)` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



I can also do the same plot by Gender and by Marital status.

```
ggplot(supermarket, aes(x = reorder_size(`State or Province`))) +
  geom_bar(aes(y = (..count../sum(..count..))) +
  xlab("State or Province") +
  scale_y_continuous(labels = scales::percent, name = "Proportion") +
  facet_grid(`Marital Status` ~ Gender) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

